# Keeps the gap.
# Leaves the brakes to the rider.

Adaptive cruise control for standing e-scooters. A camera and radar watch the road ahead. The system holds a safe distance by limiting the motor, caps speed at the legal 20 km/h, and warns the rider before a collision. It never brakes on its own.

**Exposé** for fleet operators, scooter makers and early investors

**Status** proof of concept in simulation, v0.2 **October 2026**




| time | speed | gap | time to collision | status |
|---|---|---|---|---|
| 14.5 s | 10 km/h | 4.8 m | 2.6 s | Warning |

Real data from our simulation of scenario S2: the vehicle ahead brakes at 8 s. Without rider braking, contact would follow at 17.4 s. The warning fires at 14.0 s.

## 01 The short version

**16,496**

e-scooter accidents with injuries in Germany in 2025, up 38.1 % in one year. 38 people died.

**3.4 s**

of warning before contact in our hard-braking scenario. The previous version gave none.

**25 / 25**

safety tests pass, run automatically on every change. The three safety gaps found earlier are fixed.

We have found no product that offers forward cruise control on a kick scooter. Chinese makers fit radar to e-bikes for rear warnings, and sharing operators use cameras against sidewalk riding. Our system adds forward distance keeping, with the rider always in command and an independent safety chip checking every decision.

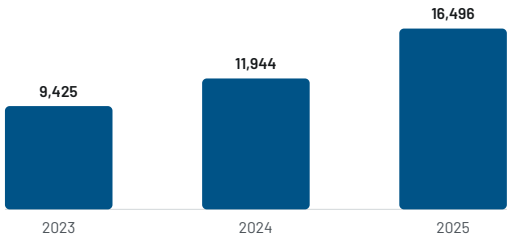
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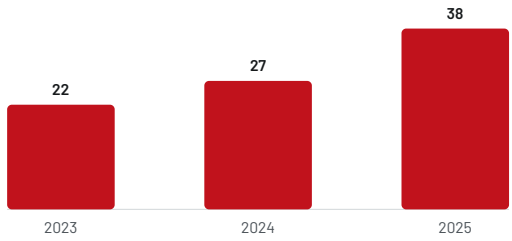
## More scooters, more crashes

Police-recorded e-scooter accidents in Germany grew 75 % from 2023 to 2025. Excessive speed and errors in dense traffic are among the leading causes.

### Accidents with personal injury




### People killed




**53.6 %**

of injured riders in 2025 were under 25.

**8.4 %**

of rider misconduct was excessive speed. 21.6 % was improper use of road or sidewalk.

**49.1 %**

of accidents happened in cities with 100,000+ inhabitants.

Source: Statistisches Bundesamt (Destatis), press releases N049 (July 2026) and N040 (July 2025).

03

# What the system does

Three behaviours, each tested in software, and one rule above them all: the rider's brake always wins.

### Keeps a safe gap

The gap grows with speed: 3 m plus 2 seconds of travel. Inside that gap the motor gets no extra power.

REQ-002 · tested

### Stays legal

Set speed is capped at 20 km/h, the German limit for e-scooters. At or above it, the motor gets no extra power.

REQ-012 · tested

### Warns early

When time to collision drops below 3 seconds, the rider hears and sees a warning. It works even after the rider has braked.

REQ-014 · tested, threshold provisional

## How it is built

### Sensors

Forward camera, FMCW radar, GNSS, IMU, wheel encoder

### Raspberry Pi 5 + AI HAT+

Object detection, radar–camera fusion, logging. Not trusted with the motor.

### STM32 safety chip

Checks every message, latches faults, enforces limits, gives the brake priority.

### Motor controller

Disabled in this version. Only through a documented manufacturer interface.

Stale sensor data, a missed heartbeat, low confidence or a rider brake removes motor authority in the same 20 ms control cycle.

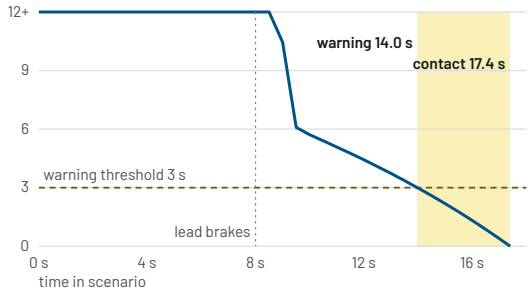
04

# Evidence, not promises

Every number here comes from code that anyone on the team can rerun. Hardware results come later, at the bench-test stage.

## The warning window

Time to collision in scenario S2. The yellow band is when the rider is warned.




## Before and after the fixes

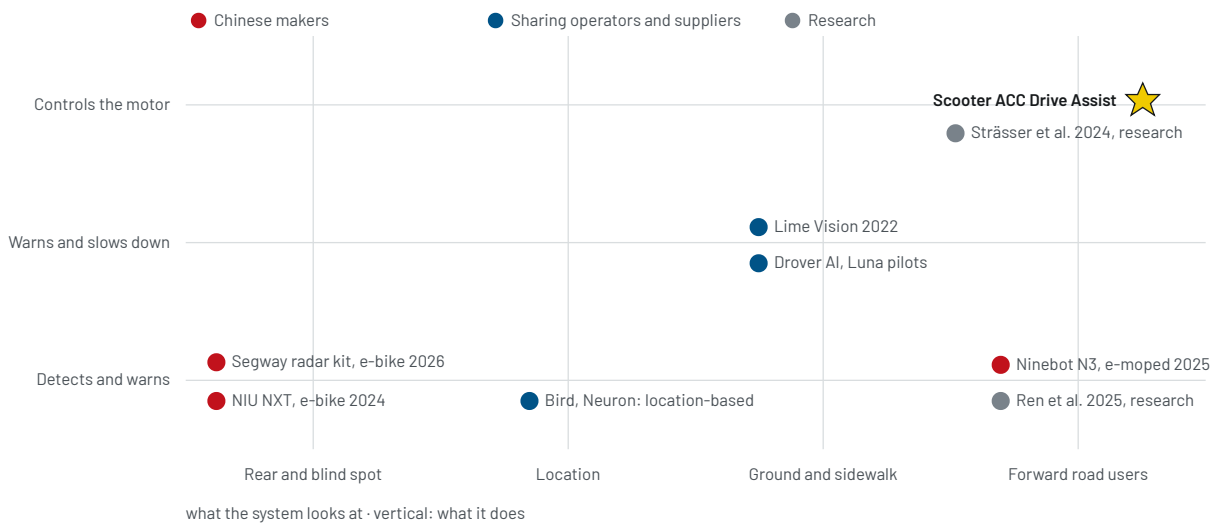
| Scenario | v0.1.0 | v0.2 today |
|---|---|---|
| Sensor fault, then rider presses brake and engage | **Accelerates again** | **Stays in safe fault mode** |
| Set speed 30 km/h on an open road | **Reaches 21.2 km/h** | **Stays below 20 km/h** |
| Vehicle ahead brakes hard | **No warning** | **Warning 3.4 s before contact** |
| Sensor data goes stale | **Motor power cut** | **Motor power cut at 10.14 s** |

Simulation with an illustrative vehicle model, not a real-world stopping test. 25 of 25 tests pass in continuous integration on Python 3.11.

05

# Where we fit

Others warn about what is behind the rider or catch riders on the sidewalk. Nobody we found keeps the gap ahead on a kick scooter.




## Why now

- Accidents are rising fast, and cities and insurers want them down.
- Edge AI has become efficient: a 2026 benchmark measured Hailo-8 as the most energy-efficient accelerator for large YOLO models on a Raspberry Pi 5.
- Chinese makers have industrialised two-wheeler radar on e-bikes since 2024, so components exist at scale.
- Sharing operators already pay for camera-based rider safety.

## Market size, handled with care

Commercial reports put micromobility in 2025 anywhere between USD 42 bn and 198 bn, and e-scooters alone at USD 47.4 bn. These numbers disagree by a factor of 4.7, so we use them as context only. The business case below is built bottom-up from units and prices you can change.

Sources: Segway press release (CES 2026); NIU Q4 2023 earnings call; The Paper, 14 Mar 2025; TechCrunch, 13 Jul 2022; Suchý & Turčaník, Sci. Rep. 2026; IMARC and Grand View Research listings, accessed 17 Oct 2026.

06

# Business model

We start where safety already has a budget: shared fleets in the EU. Germany first, because its rules are clear and strict.

### Sharing operators

A retrofit kit plus a yearly software and data service. Fewer crashes and less damage, proof of speed compliance for city permits.

first customers

### Scooter makers

License the controller, safety-chip logic and test evidence. The maker builds the hardware at volume, often in China.

scale

### Insurers and cities

Anonymised warning and near-miss statistics for risk pricing and hot-spot maps. No raw video leaves the scooter.

later

## Getting into the EU market

| Requirement | What applies | Our status |
|---|---|---|
| Vehicle rules, Germany | eKFV: 6–20 km/h, operating permit (ABE) and insurance plate | **20 km/h cap done in software. Permit impact to assess.** |
| Radio and CE | Radio Equipment Directive 2014/53/EU for the radar | **Module selection** |
| Battery | Regulation (EU) 2023/1542, battery passport from Feb 2027 | **Scooter maker's duty** |
| Camera data | GDPR (EU) 2016/679 | **Processing on the device, no raw video stored by default** |
| Product law | GPSR 2023/988, Product Liability Directive 2024/2853, AI Act 2024/1689 | **Legal assessment before road tests** |

China supplies technology and partners, not customers: kick scooters are banned from public roads in Beijing and other major cities (Beijing rules reaffirmed from 1 May 2026).

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07

# Profit model

A scenario, not a forecast. Move the sliders to test your own assumptions. Every default is a placeholder until supplier quotes and operator interviews replace it.

**Kit price per scooter** — €420

Retrofit kit sold to a fleet operator.

**Kit cost per scooter at volume** — €260

Target with a custom board. The prototype parts list costs €715–1,420.

**Software and data service per scooter and year** — €36

Billed on every installed kit.

**Fixed costs per year** — €650 k

Team, testing, certification, in thousand €.

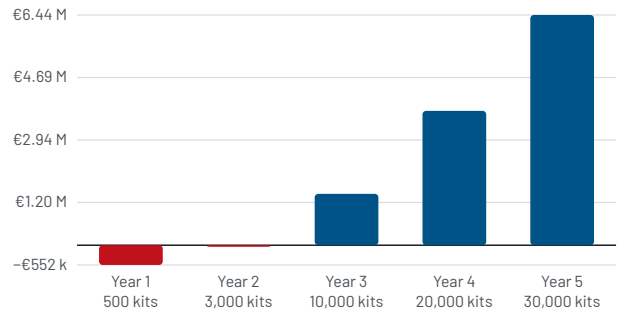
**Volume ramp** — 100 %

100 % = 500, 3,000, 10,000, 20,000 and 30,000 kits in years 1–5.

**€160**
gross margin per kit

**4,063**
kits per year to break even

**€6.44 M**
operating result in year 5




Model: result = kits sold × (price − cost) + installed kits × service − fixed costs. Installed kits accumulate with no attrition. Excludes taxes, financing and working capital. These are illustrative assumptions to be tested in hypotheses B1 (willingness to pay) and B3 (cost-down).

## 08 Impact

The goal is fewer injuries without taking control away from the person on the scooter.

### Safety

Distance keeping and early warnings address speed and reaction errors in dense traffic. The real effect has to be measured in a pilot with a control fleet. We will not claim a crash reduction before that.

### Rider authority

No automatic braking, because a standing rider can lose balance. The system only withholds extra power and warns. Brake and cancel always win.

### Young riders

More than half of injured riders are under 25. A speed cap and early warnings help the least experienced riders most.

### Privacy

Images are processed on the scooter. Only object lists and statistics leave it, so city and insurer data contain no faces.

## 09 Scope and the road to production

Each stage ends at a gate with evidence. We are at stage one.

| 1 Concept | 2 Lab | 3 Field | 4 Pilot | 5 Industrialise | 6 Series |
|---|---|---|---|---|---|
| Simulation, tests, requirements | Safety chip, emulated motor, fault injection | Closed test track only | Custom board, small fleet with a partner | Design for manufacture, end-of-line test | Production and field monitoring |
| now | gate: timing measured | gate: safety review + site approval | gate: reliability data | gate: yield | gate: field KPIs |

### In scope now

Control logic, simulation, safety requirements and tests, sensor-data

### Out of scope now

Real motor commands, automatic braking, public-road tests and certification. These need a documented safety review first.

replay, bench test with an emulated motor, offline model training.

## 10 What we are looking for

### A pilot fleet

A sharing operator in Germany willing to run a measured pilot with a control group.

### A hardware partner

A scooter maker or ODM with eKFV-approved models and a documented motor-controller interface.

### Early funding

For the lab and field stages. The amount will be set with the stage-2 budget.

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> This exposé describes a research prototype in simulation. It makes no claim of road legality, ISO compliance, certified safety or validated stopping performance. Market figures are third-party estimates; the profit model is an illustrative scenario.

## Sources


1. Destatis, "38,1 % mehr E-Scooter-Unfälle mit Personenschaden im Jahr 2025," press release N049, Jul. 2026.
2. Destatis, "26,7 % mehr E-Scooter-Unfälle mit Personenschaden im Jahr 2024," press release N040, Jul. 2025.
3. eKFV (Elektrokleinstfahrzeuge-Verordnung), §1 and §2, gesetze-im-internet.de.
4. Segway, radar kit for e-bikes, PR Newswire, Jan. 2026.
5. NIU Technologies, Q4 2023 earnings call (NXT e-bike with rear mmWave radar).
6. The Paper (澎湃新闻), Ninebot and NIU smart riding features, 14 Mar 2025.
7. TechCrunch, "Lime has built its own camera-based sidewalk detection technology," 13 Jul 2022.
8. China Daily, "Beijing mandates helmets for e-bike users, bans scooters," 7 Jan 2026.
9. I. Suchý and M. Turčaník, energy efficiency of YOLOv8 and RT-DETR on edge devices, Sci. Rep. 16, 10908, 2026.
10. IMARC Group and Grand View Research market listings, accessed 17 Oct 2026.
11. EU legal acts: 2014/53/EU, (EU) 2023/1542, (EU) 2016/679, (EU) 2023/988, (EU) 2024/2853, (EU) 2024/1689.
12. Project evidence: repository karimsamy11/scooter-acc-drive-assist-v2 (private), PROJECT_DOCUMENT and MARKET_CHINA_EU.